

Sales Forecasting Across Multiple Retail Stores

Technical Blog Report - Rossmann Pharmaceuticals Project

1. Executive Summary & Business Need

Rossmann Pharmaceuticals required an automated, data-driven solution to forecast daily store sales across 1,115 retail locations up to 6 weeks ahead. Replacing manual estimations with end-to-end machine learning ensures optimized inventory allocation and marketing planning.

2. Exploratory Data Analysis Key Findings

- Promotional Impact: Active daily promos boosted turnover by +38.4% with high customer spikes.
- Holiday Seasonality: Pre-holiday stocking trends drive significant demand spikes before closure.
- Competitor Proximity: High-traffic urban locations offset close competitor presence.

3. Machine Learning & Preprocessing Pipeline

Data preprocessing inside `src/data_processing.py` extracted temporal features (`IsWeekend`, `Month`, `WeekOfYear`) and handled missing values. A `RandomForestRegressor` pipeline was serialized (`.pkl`).

Model Architecture	Validation MAE (\$)	Validation RMSE (\$)
Random Forest Regressor	1376.34	1919.47

4. Deep Learning Approach (Stacked LSTM)

Stationarity was verified using Augmented Dickey-Fuller (ADF) testing. First-order differencing and a 6-week sliding window sequence were used to train a two-layer stacked LSTM network.

5. Deployment & Final Project Deliverables

- Streamlit Web Dashboard: Interactive frontend serving model inference in `app/main.py`
- GitHub Repository: <https://github.com/abhivagadiya1600-lgtm/tellco-analytics>